

Worklet Details

1. Worklet ID: 25ST31BMS
2. College Name: B.M.S. College of Engineering

KPIs achieved till now

- **Dataset Generation:** Successfully curated and balanced a custom 5,000-clip dataset with a strict 70/15/15 split.
- **Acoustic Mixing Validation:** Validated dataset difficulty using a frozen YAMNet baseline, achieving an F1 score of ~0.80, proving the background noise mixing is effectively challenging.
- **Model Benchmarks:** Achieved an F1 score of 0.9305 and AUC of 0.9807 using an Audio Spectrogram Transformer (AST).
- **Ensemble Testing:** Tuned a late-fusion ensemble (AST + CNN14) to reach a peak F1 of 0.9332.

Next Steps

- Train the highly-optimized EfficientAT MN10-AS model to establish our edge-deployment baseline for SmartThings compatibility.
- Execute zero-shot text-guided classification using the CLAP model.
- Begin compiling the final 4-model team ensemble (E20) targeting an F1 score of >0.96.
- Run the "Whisper Feature Mode" novelty experiment to test robustness against human speech background noise.

Any Challenges/ Issues faced

- **Data Leakage/Difficulty Tuning:** Ensuring the mixed background noise wasn't completely masked by the cooking sounds. (Overcome by testing against the YAMNet baseline).
- **Compute Limitations:** Managing VRAM constraints on Google Colab when fine-tuning heavy Transformer models like AST (87M parameters).

Key Achievements/ Outcome till now

- Successfully simulated real-world kitchen environments by mixing 2,500 foreground cooking events with complex background noise (TV, speech, washing machines) at varying SNRs.
- Established a definitive architectural baseline proving that Transformer models (AST) significantly outperform standard CNNs (CNN14: 0.82 F1) on this specific acoustic dataset.

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Last Month's Progress:

- Completed programmatic generation of the **kitpri/** dataset, yielding 5,000 strictly balanced audio clips (16kHz, 5s, mono).
- Successfully implemented **Strategy B (Noise-Robust Training)** by mixing foreground cooking events with complex background audio (MUSAN, UrbanSound8K) at varying Signal-to-Noise Ratios (SNRs).
- Extracted dense acoustic embeddings across all dataset splits using pre-trained backbones to accelerate downstream head training.

KPIs Achieved as per Milestone:

- **Data Engineering:** Finalized and locked a strict 70/15/15 stratified split for fair model evaluation.
- **Baseline Establishment:** Deployed a frozen YAMNet architecture to establish our minimum viability threshold.
- **Core Experiment:** Successfully executed the primary CNN vs. Transformer comparative analysis (CNN14 vs. AST)

Results Till Now (Evaluated on kitpri/ Test Split):

- Baseline (YAMNet): ~0.80 F1 Score.
- CNN14 (PANNs): 0.8235 F1 | 0.8861 AUC.
- Audio Spectrogram Transformer (AST): **0.9305 F1 | 0.9807 AUC.**
- Ensemble (AST + CNN14 Tuned): **0.9332 F1.**

Observations Based on Results:

- **Transformers Dominate:** The ViT-based AST model provided a massive ~11% F1 improvement over the CNN14 baseline. The Transformer's ability to capture long-range temporal context makes it significantly better at separating cooking transients from continuous background noise.
- **Ensembles Push the Ceiling:** Late-fusion ensembling (combining the local feature extraction of CNNs with the global context of Transformers) effectively minimized false positives, achieving our highest overall accuracy metric.

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Resource Requirement Identified with Status

- **Requirement:** Cloud GPU compute (Nvidia T4 or higher) with at least 12GB RAM for training AST, CLAP, and potential BEATs models.
- **Status:** Currently utilizing Google Colab (Free Tier / T4) combined with Google Drive for shared team dataset hosting.

Github Repository Link and PRISM Portal Registration Status

State: "Registered on PRISM portal and tracking progress"

Any Anticipated Break (Due to exams etc)

Data Collection Information

- **Is the data self-collected?** No.
- **Details:** The `kitpri/` dataset was programmatically synthesized using open-source audio datasets.
- **Sources:** Audio clips were sourced from ESC-50, FSD50K, and AudioSet (for primary cooking events), aggressively mixed with UrbanSound8K and MUSAN to provide realistic, challenging background noise augmentation.

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Professor Comments (To be filled in and presented by professor (Either)

Status and future plan
Latest Literature Review
Expert Insights
Any other comment

COMPUTATION RESOURCE

Details:

- **Google Colab:** Used extensively by all 4 team members.
- **CPU Instances:** Used for lightweight feature extraction (frozen YAMNet / VGGish) and dataset preprocessing (~20-40 min sessions).
- **GPU Instances (T4):** Required for CNN14 and AST fine-tuning. Averaging 2 to 4 compute hours per model training session.